INCH-POUND
MIL-W-16878/16B
11 September 1992
SUPERSEDING
MIL-W-16878/16A(NAVY)
10 August 1981

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, CROSSLINKED, MODIFIED POLYETHYLENE (XLPE) INSULATED, 125°C, 3000 VOLTS

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-16878.

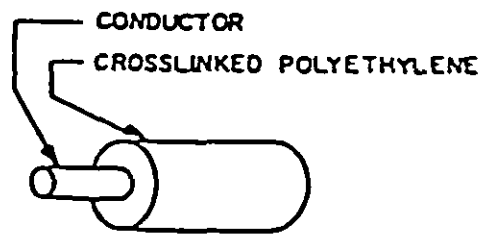


FIGURE 1. Construction.

AMSC N/A FSC 6145
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TABLE I. Construction details.

PIN 1/	Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material	Coating		Min	Max
M16878/16-BDA*	26	1 X 26	Copper	Tin	0.0159	0.067	0.082
M16878/16-BDB*	26	7 X 34	Copper	Tin	.019	.070	.085
M16878/16-BDE*	26	19 X 38	Copper	Tin	.020	.070	.085
M16878/16-BEA*	24	1 X 24	Copper	Tin	.0201	.071	.086
M16878/16-BEB*	24	7 X 32	Copper	Tin	.024	.075	.090
M16878/16-BEE*	24	19 X 36	Copper	Tin	.025	.075	.090
M16878/16-BFA*	22	1 X 22	Copper	Tin	.0254	.076	.091
M16878/16-BFB*	22	7 X 30	Copper	Tin	.030	.081	.098
M16878/16-BFE*	22	19 X 34	Copper	Tin	.032	.081	.098
M16878/16-BGA*	20	1 X 20	Copper	Tin	.0320	.083	.098
M16878/16-BGB*	20	7 X 28	Copper	Tin	.038	.089	.105
M16878/16-BGE*	20	19 X 32	Copper	Tin	.040	.089	.105
M16878/16-BHA*	18	1 X 18	Copper	Tin	.0403	.091	.106
M16878/16-BHB*	18	7 X 26	Copper	Tin	.048	.099	.114
M16878/16-BHE*	18	19 X 30	Copper	Tin	.047	.099	.114
M16878/16-BJA*	16	1 X 16	Copper	Tin	.0508	.102	.117
M16878/16-BJE*	16	19 X 29	Copper	Tin	.057	.108	.123
M16878/16-BJF*	16	26 X 30	Copper	Tin	.060	.111	.125
M16878/16-BKA*	14	1 X 14	Copper	Tin	.0641	.115	.130
M16878/16-BKE*	14	19 X 27	Copper	Tin	.072	.123	.138
M16878/16-BKH*	14	41 X 30	Copper	Tin	.076	.127	.142
M16878/16-BLE*	12	19 X 25	Copper	Tin	.092	.156	.171
M16878/16-BLG*	12	37 X 28	Copper	Tin	.089	.153	.168
M16878/16-BLJ*	12	65 X 30	Copper	Tin	.093	.157	.172
M16878/16-BMG*	10	37 X 26	Copper	Tin	.111	.174	.189
M16878/16-BNL*	8	133 X 29	Copper	Tin	.167	.239	.254
M16878/16-BNM*	8	168 X 30	Copper	Tin	.169	.239	.254
M16878/16-BPL*	6	133 X 27	Copper	Tin	.210	.283	.298
M16878/16-BRL*	4	133 X 25	Copper	Tin	.268	.346	.366
M16878/16-BSL*	2	133 X 23	Copper	Tin	.355	.415	.435
M16878/16-BSP*	2	665 X 30	Copper	Tin	.342	.415	.435
M16878/16-BTN*	1	259 X 25	Copper	Tin	.378	.468	.488
M16878/16-BTR*	1	817 X 30	Copper	Tin	.382	.468	.488
M16878/16-BUN*	0	259 X 24	Copper	Tin	.424	.518	.538
M16878/16-BUS*	0	1045 X 30	Copper	Tin	.431	.518	.538
M16878/16-BWT*	00	1330 X 30	Copper	Tin	.475	.575	.599
M16878/16-BYV*	000	1672 X 30	Copper	Tin	.535	.657	.681
M16878/16-BZW*	0000	2109 X 30	Copper	Tin	.600	.709	.733

1/ PIN stands for part or identifying number.

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ADDITIONAL REQUIREMENTS:

Visual and mechanical examination: Required.
Spark test: 8.0 kV.
Impulse dielectric test: 12.0 kV.
Dielectric withstanding voltage: 6.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$.
Partial discharge test: Required per T-24-380 of Insulated Cable Engineers Association.

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 10,000.
D = Maximum average diameter of finished wire.
d = Conductor diameter.

Cold bend: Condition 4 hours at minus 55 ± 1°C (see table II).

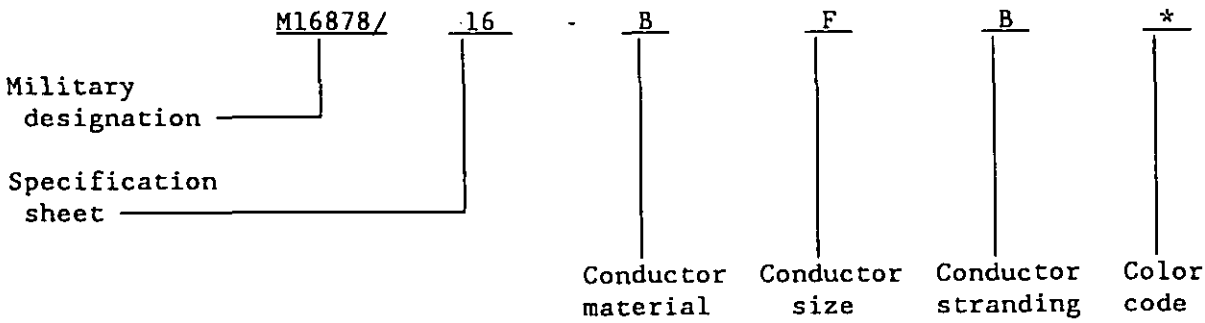
TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter in inches)
26 through 22	2
20 through 14	3
12, 10	4.5
8	6
6, 4	10
2, 1, 0	18
00, 000, 0000	20

Concentricity: 70 percent (minimum).
Surface resistance: Not required.
Wrap back: Not required.
Heat resistance: Condition at 150°C.
Flammability: Meets requirements of UL VW1 Flame Test.
Toxicity: Meets requirements of concentration levels specified in MIL-E-917.
Shrinkage: Required.
Heat aging: 25 percent change (maximum) in 96 hours at 135°C.
Insulation tensile strength: 1800 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Marking and stripe durability: Required.
Fungus resistance: Required.
Maximum dc resistance of finished wire: See table I of MIL-W-16878.
Identification of product: Required for wire sizes 22 AWG and larger.

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PIN example (see MIL-W-16878):



Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6145-N330-15)

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